The Connected River (Part 2) - Exploring

Using integrated monitoring evidence to understand river health and change

This work was carried out as part of the UK Environmental Exposure Hub (EEx Hub), developed through the UKCEH National Capability for UK Challenges (NC-UK) programme and funded by NERC (NE/Y006208/1).

What this notebook explores

Environmental monitoring provides different pieces of evidence about river systems. Part 1 connects those observations through space and time. This part uses the finished dataset to ask:

- where do we have enough overlapping evidence?
- how has biological monitoring changed through time?
- how is river biology associated with environmental pressures?
- what do repeated observations show that spatial comparisons cannot?
- what does the longer macroinvertebrate record show about ecological change?

The figures are descriptive. They demonstrate the questions enabled by the integration, not causal effects of individual pressures.

Open the connected dataset

Run The Connected River (Part 1) - Building first. By default this notebook reads the most recent output recorded in `connected_river_workspace/latest_method_run.json`. Set `BCR_RELEASE_DIR` only when opening an output stored somewhere else.

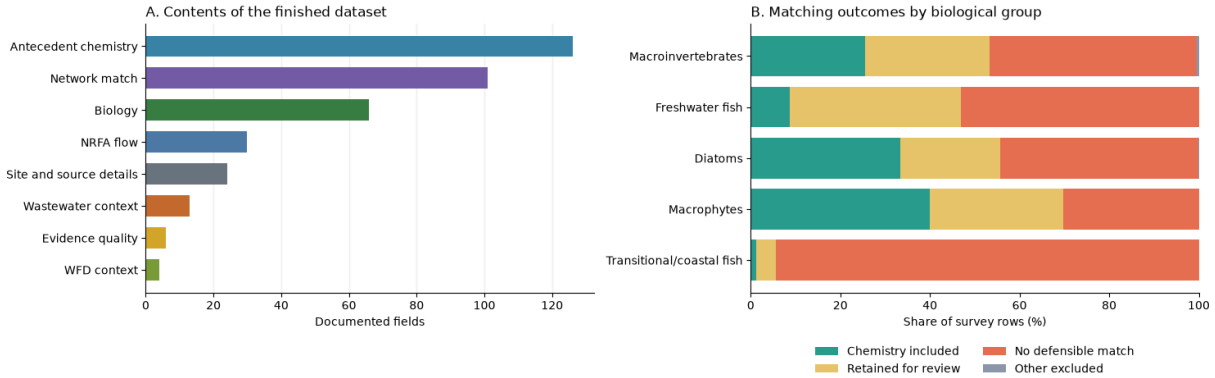
The open ChemPop v2 archive is used only for the final historical continuation. Supply an existing ZIP with `BCR_HISTORICAL_ARCHIVE`, or leave it blank to download and cache the 43 MB deposit.

Meet the connected dataset

Each row begins with a biological survey. Where the matching checks support it, the row also carries antecedent chemistry, route evidence, flow, wastewater and regulatory context. See The Connected River (Part 1) - Building for how those connections were constructed and evaluated.

In [2]:
```
dataset_figure, assemblage_summary = plot_dataset_overview(
    specialist_path,
    completed.dictionary,
    output_directory=output_directory,
)
plt.show()
assemblage_display = assemblage_summary[[
    "survey rows", "biology sites", "chemistry included (%)"
]].round(1)
for count_column in ("survey rows", "biology sites"):
    assemblage_display[count_column] = assemblage_display[count_column].map(
        lambda value: f"{int(value):,}"
    )
display(Markdown(dataframe_markdown(assemblage_display)))
```

53,256 surveys · 12,234 biological sites · 2018-2025 · 370 documented fields



biological group	survey rows	biology sites	chemistry included (%)
Macroinvertebrates	26,015	5,953	25.5
Freshwater fish	12,262	5,027	8.8
Diatoms	8,038	2,530	33.3
Macrophytes	5,290	3,183	40
Transitional/coastal fish	1,651	343	1.2

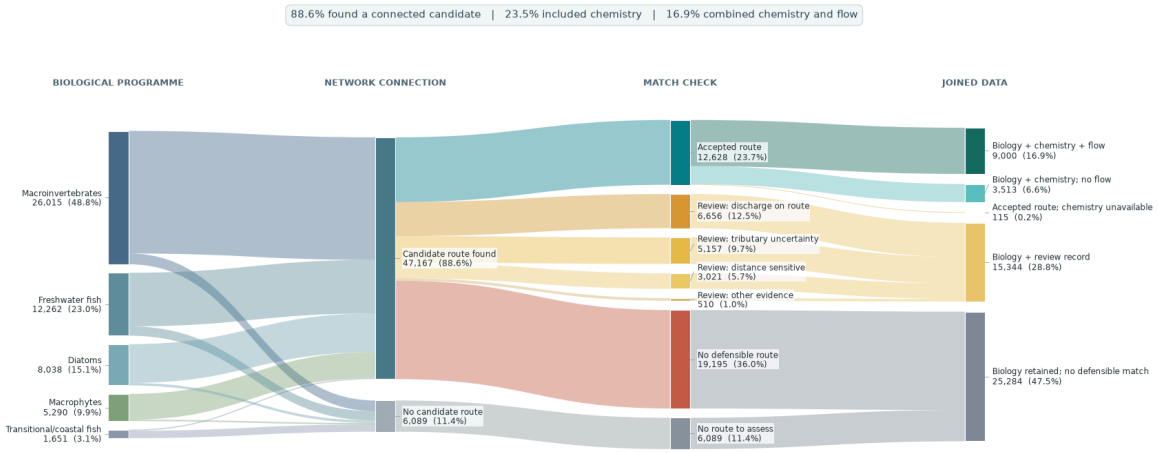
From biological surveys to integrated evidence

The diagram follows biological survey rows from candidate routes through the matching checks and into the joined dataset. Review and unmatched rows remain in the saved matching tables.

In [3]:
```
flow_columns = [
    "data_origin", "assemblage", "candidate_count", "match_decision",
    "chemistry_included_in_release", "flow_evidence_available",
]
flow_records = pd.read_parquet(specialist_path, columns=flow_columns)
flow_records = flow_records.loc[flow_records["data_origin"].eq("connected_ri
evidence_figure, evidence_summary = plot_evidence_river(flow_records)
evidence_figure.savefig(
```

```
    output_directory / "joined_evidence_sankey.png", dpi=180, bbox_inches="t
)
plt.show()
```

Survey records through the matching pipeline (53,256 surveys)



Band widths are survey records. Review and unmatched rows remain in the saved matching tables.

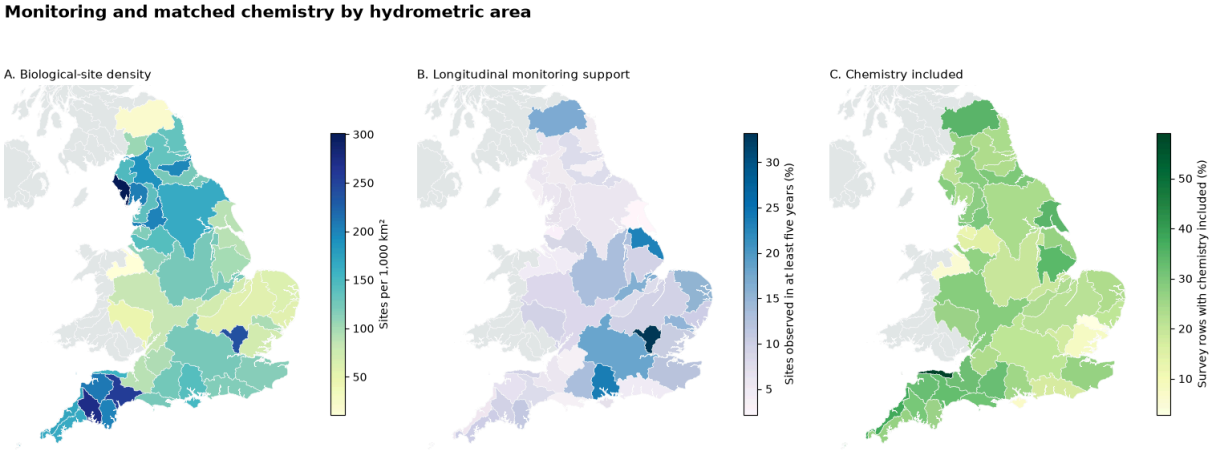
Where do we have evidence?

The first step is to establish where the available evidence supports analysis.

National monitoring and matching support

The maps separate the biological monitoring network, repeat-site coverage and included chemistry. The usable evidence is determined by their overlap, not by the size of any one source dataset.

In [4]:
```
spatial = prepare_spatial_evidence(RELEASE_DIR, specialist_path)
spatial_figure, spatial_summary = plot_spatial_support(
    spatial, output_directory=output_directory
)
plt.show()
lowest_support = spatial_summary.nsmallest(3, "chemistry included rows (%)")
display(Markdown(
    "**Lowest included-chemistry coverage in monitored areas:** " + "; ".joi
        f"{row['hydrometric area']} ({row['chemistry included rows (%)']:.1f
        for _, row in lowest_support.iterrows()
    )
))
```

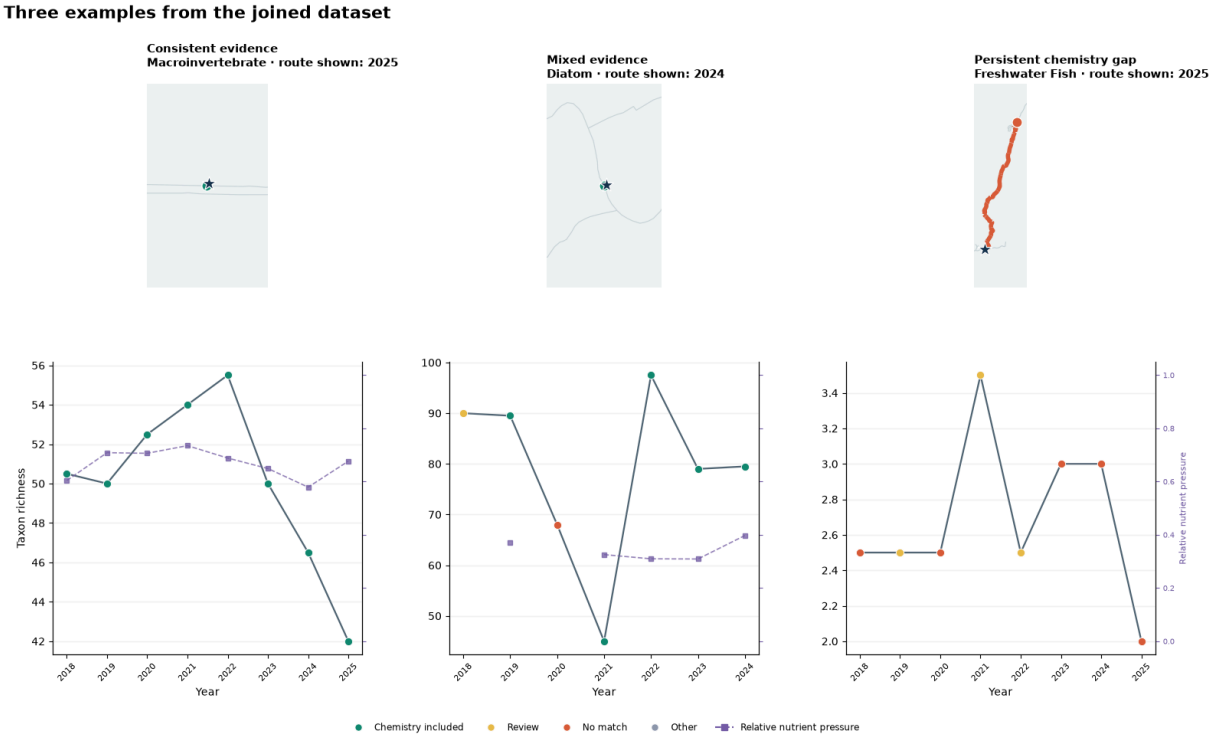


Lowest included-chemistry coverage in monitored areas: Stour (Essex and Suffolk) (2.7%); Dee (Cheshire) (5.3%); Isle of Wight (7.5%)

Three river stories

These examples show three common situations: consistently supported evidence, changing evidence between years and a persistent chemistry gap. They were selected using completeness and match category.

In [5]:
```
story_figure = plot_river_stories(
    specialist_path, RELEASE_DIR, spatial,
    output_directory=output_directory,
)
plt.show()
```



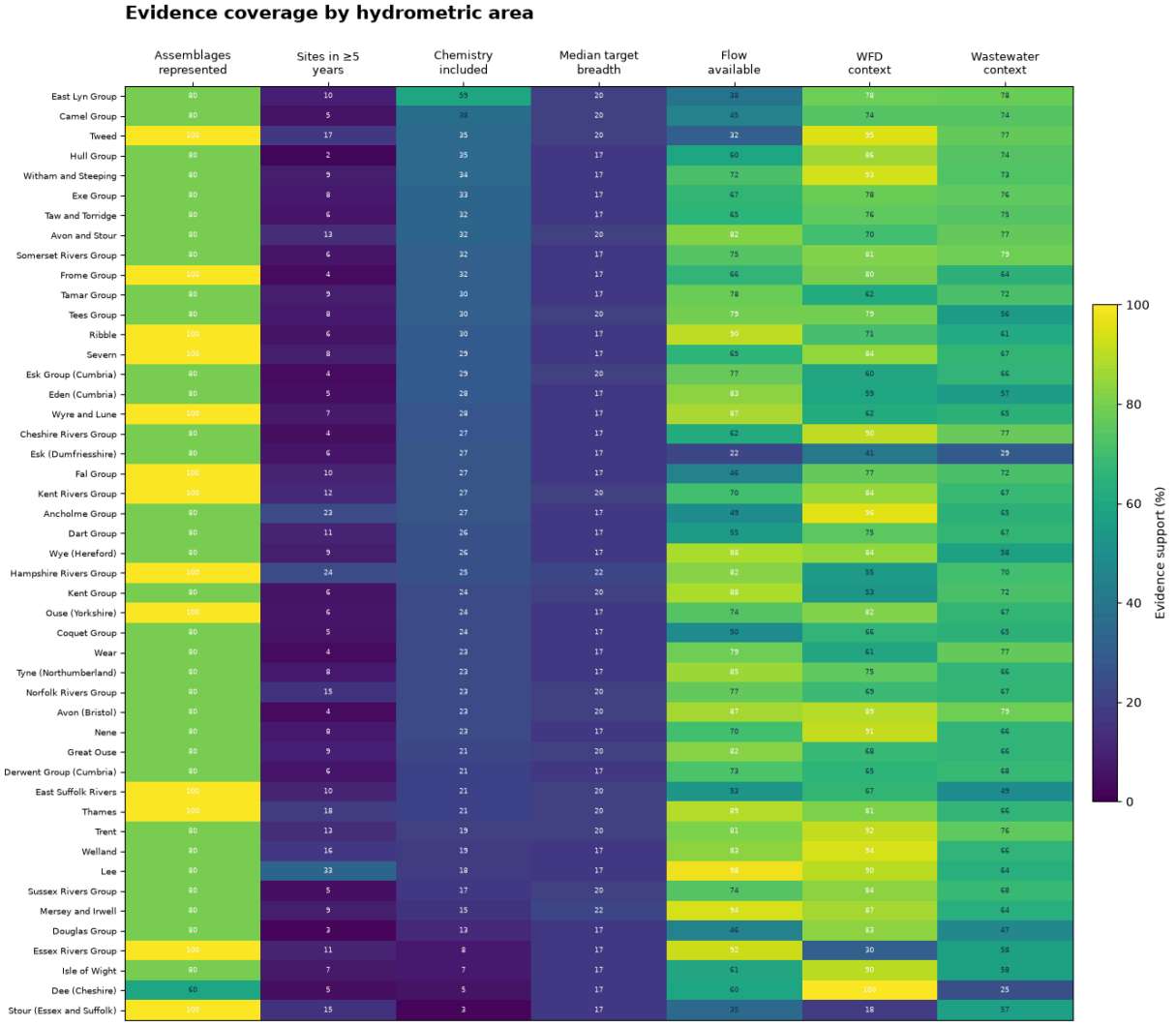
Where are the evidence gaps?

Integrated evidence by hydrometric area

The fingerprint shows which kinds of analysis are supported in different places. Each column uses a common 0–100 availability scale; it is not an overall river-health score. Rows are ordered by included chemistry and repeat-site support.

In [6]:
```
fingerprint_figure, fingerprints = plot_evidence_fingerprint(
    specialist_path, spatial, output_directory=output_directory
)
plt.show()

fingerprint_preview = fingerprints[[
    "catchment_name", "chemistry included", "repeat-site support",
    "flow available", "WFD context", "wastewater context",
]].head(6).round(1)
display(Markdown(dataframe_markdown(fingerprint_preview, include_index=False
```

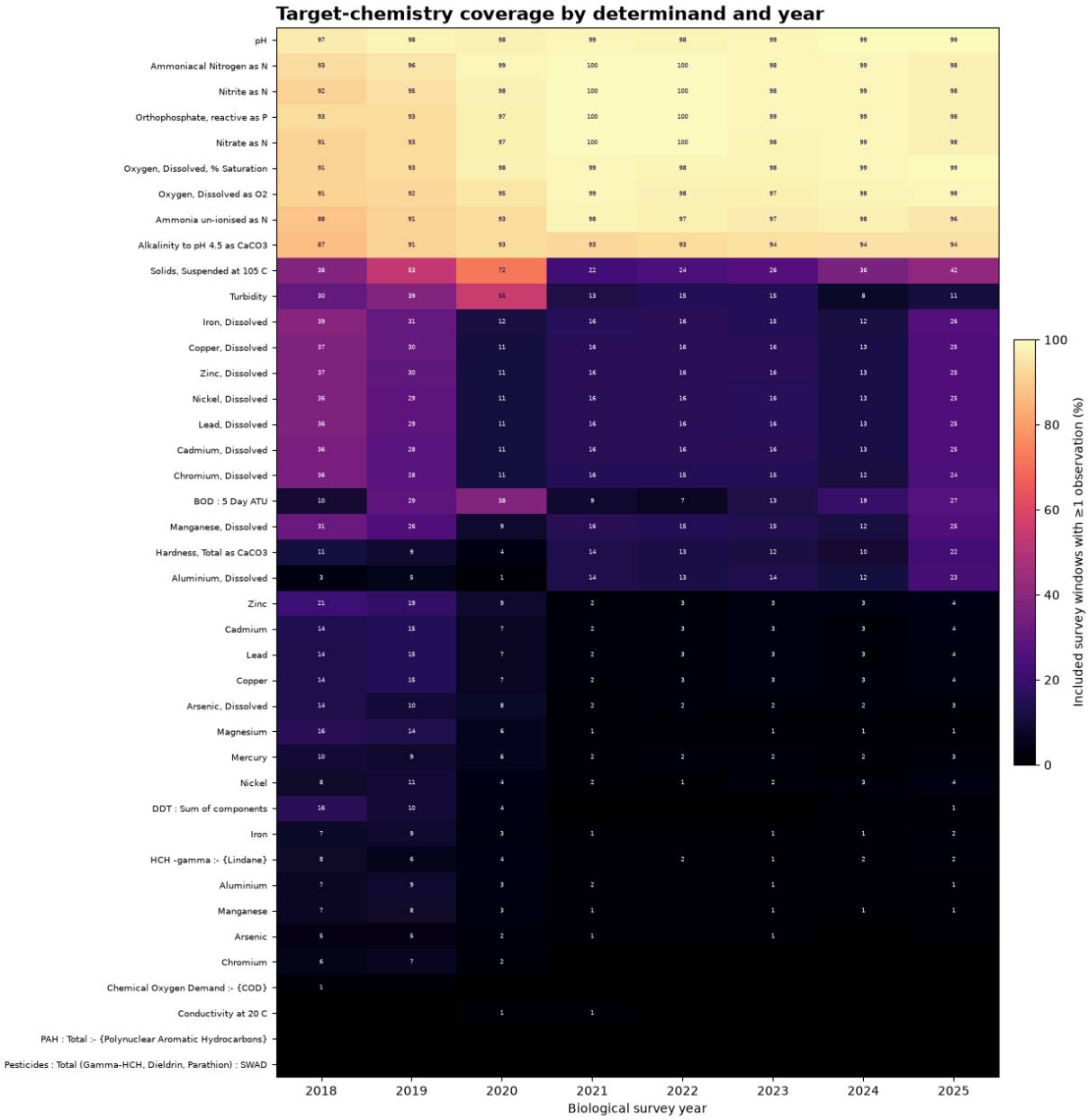


catchment_name	chemistry included	repeat-site support	flow available	WFD context	wastewater context
East Lyn Group	59.1	9.6	38.4	78.3	77.8
Camel Group	37.6	5.1	45.4	73.9	74.3
Tweed	34.7	16.7	31.5	94.9	76.7
Hull Group	34.7	2.1	60	86.2	74.2
Witham and Steeping	34.3	9	72.1	93.1	73.3
Exe Group	33	7.8	66.9	77.8	75.8

Chemistry coverage through time

Not every determinand is measured in every survey window. The heatmap shows that coverage directly, with missing measurements left as missing.

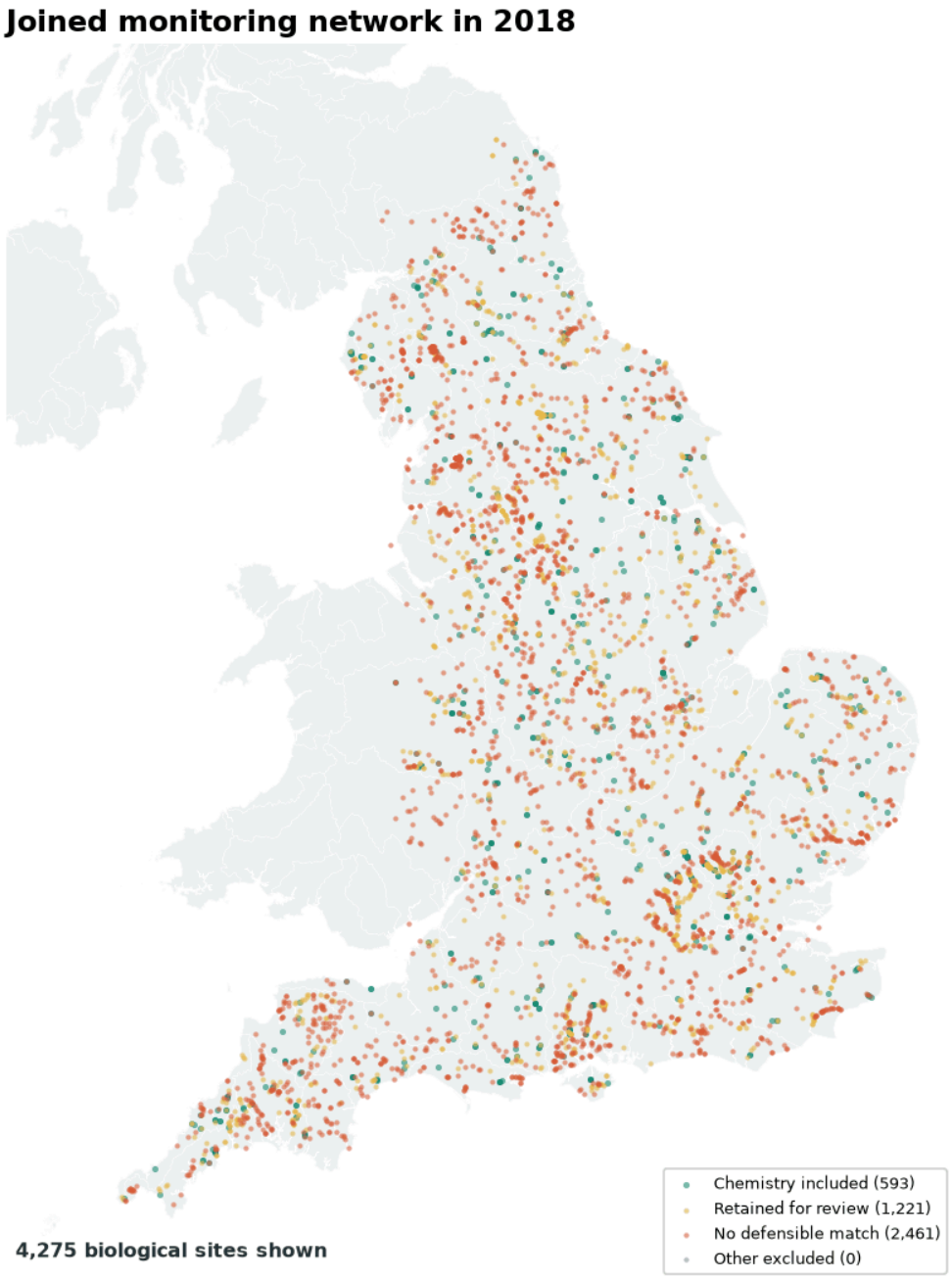
In [7]:
```
chemistry_figure, chemistry_coverage = plot_chemistry_coverage(
    specialist_path, output_directory=output_directory
)
plt.show()
```



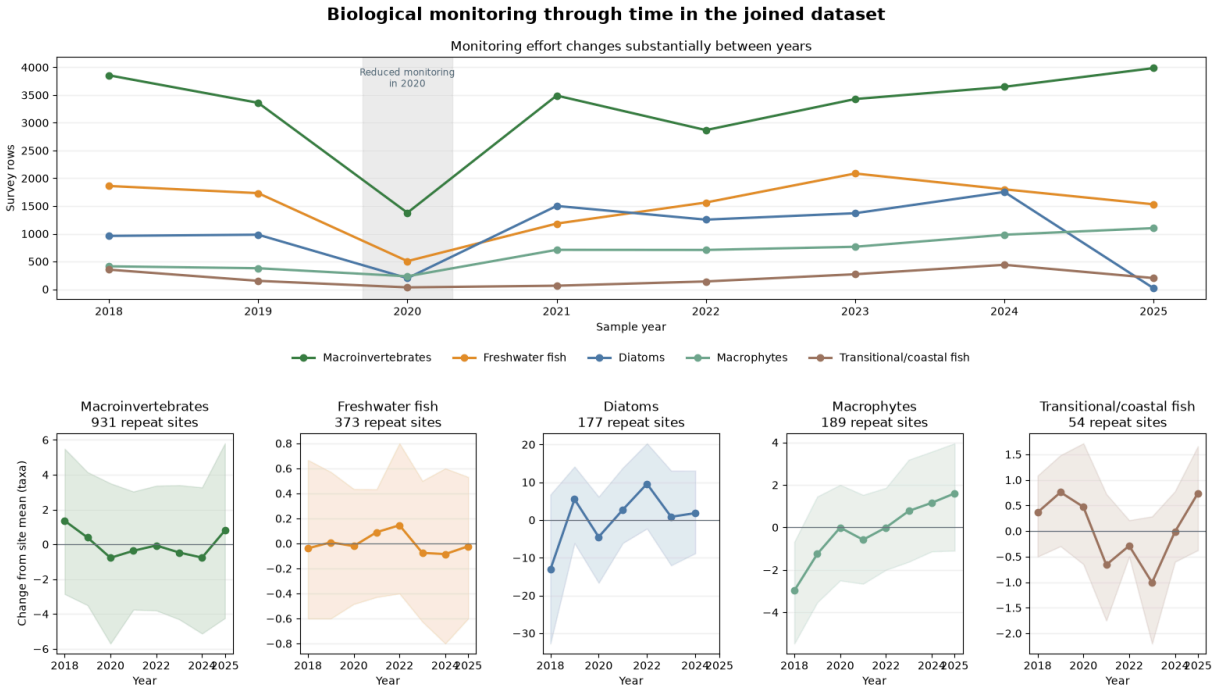
Monitoring through time

When the notebook is run, the first map is interactive: zoom, pan and hover over sites to inspect the 2025 network. Major river links are drawn behind the monitoring sites, so dense clusters can be separated at local scale. The stored animation gives the annual overview, and the following figure separates changing survey effort from trends at repeat sites.

Monitoring fell sharply across programmes in 2020. That interruption coincides with COVID-19 restrictions and should be treated as a sampling-effort break, not as evidence of sudden ecological change. The latest year can also be incomplete for individual programmes, so trend interpretation uses the saved coverage counts alongside the biological values.



In [9]:
```
biology_figure, trend_summary = plot_biological_monitoring(
    specialist_path, RELEASE_DIR, output_directory=output_directory
)
plt.show()
trend_display = trend_summary.rename(columns={
    "repeated_sites": "sites", "median_slope": "median slope",
    "lower_quartile": "lower quartile",
    "upper_quartile": "upper quartile",
}).round(2)
trend_display.index.name = "biological group"
display(Markdown(dataframe_markdown(trend_display)))
```



biological group	sites	median slope	lower quartile	upper quartile
Macroinvertebrates	931	-0.14	-1.1	0.93
Freshwater fish	373	-0	-0.15	0.11
Diatoms	177	2	-1.27	6.29
Macrophytes	189	0.6	-0.08	1.2
Transitional/coastal fish	54	-0.02	-0.26	0.09

From monitoring to environmental relationships

Spatial pattern versus repeat-site change

The two rows answer different questions. The top row compares site medians and shows the national spatial pattern. The bottom row centres both variables within repeat sites and looks at changes through time at the same site.

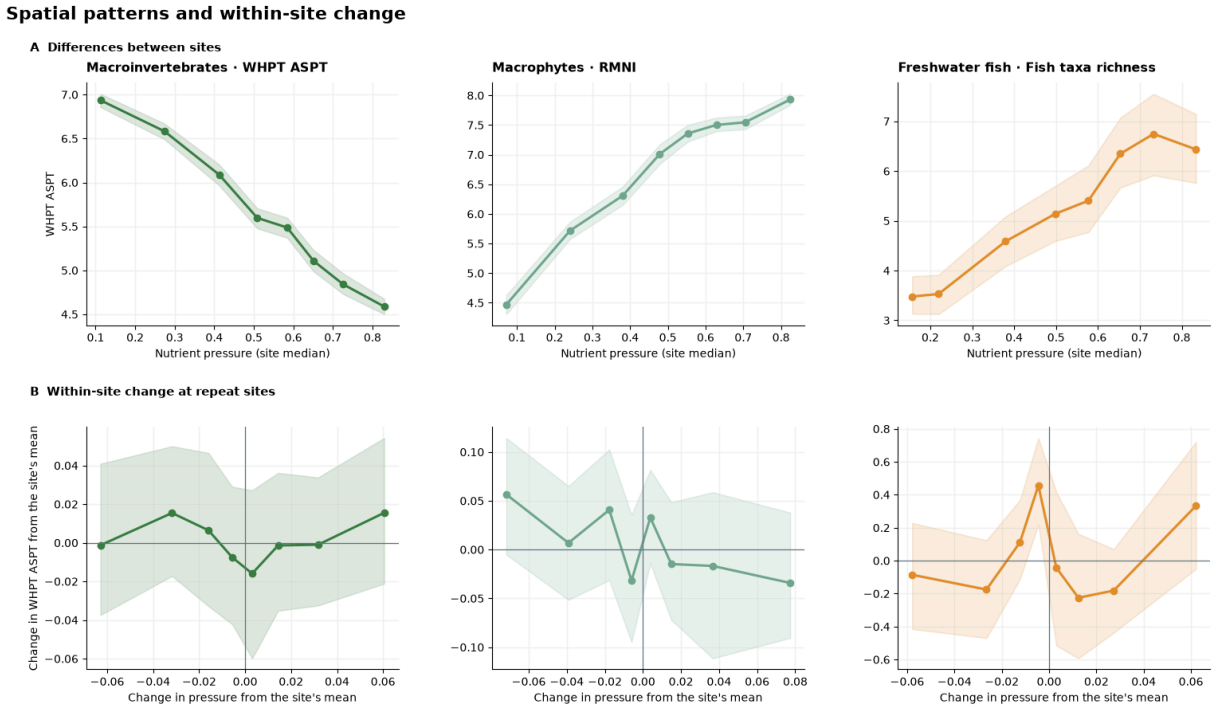
The outcomes are WHPT ASPT for macroinvertebrates, RMNI for macrophytes and taxa richness for freshwater fish. Points are site-balanced pressure bins and bands are bootstrap intervals. These are descriptive plots and should not be read as causal effects.

In [10]:
```
response_figure, response_summary, response_curves = plot_pressure_response_
    specialist_path, output_directory=output_directory
)
plt.show()

response_display = response_summary.copy()
response_display.insert(0, "measure", response_display.pop("outcome"))
response_display = response_display.drop(columns=["assemblage", "pressure"])
```

```
    "sites in spatial curve": "spatial sites",
    "spatial Spearman correlation": "spatial rho",
    "sites in repeat-site curve": "repeat sites",
    "within-site Spearman correlation": "within-site rho",
})
for count_column in ("spatial sites", "repeat sites"):
    response_display[count_column] = response_display[count_column].map(
        lambda value: f"{int(value):,}"
    )
response_display[["spatial rho", "within-site rho"]] = response_display[[
    "spatial rho", "within-site rho"
]].round(2)
display(Markdown(dataframe_markdown(response_display, include_index=False)))

display(Markdown(
    "**Reading the curves.** Macroinvertebrate WHPT ASPT falls and macrophyt
    "rises across the national nutrient gradient, while fish richness also r
    "The repeat-site curves are much flatter. The positive pooled fish relat
    "probably reflects differences in habitat or river size between sites ra
    "a beneficial nutrient response."
))
```



measure	spatial sites	spatial rho	repeat sites	within-site rho
WHPT ASPT	1,982	-0.71	270	-0.03
RMNI	1,405	0.72	101	-0.13
Fish taxa richness	563	0.41	82	0.02

Reading the curves. Macroinvertebrate WHPT ASPT falls and macrophyte RMNI rises across the national nutrient gradient, while fish richness also rises. The repeat-site curves are much flatter. The positive pooled fish relationship probably reflects differences in habitat or river size between sites rather than a beneficial nutrient response.

> **SPATIAL ASSOCIATION ≠ TEMPORAL RESPONSE**
> Strong differences between sites can reflect river size, habitat, region or monitoring design. They do not show that changing one pressure will produce the same response through time at a particular site.

What this analysis does not establish

A formal model would also need to account for site, year, season, flow, sampling method, nonlinear relationships, spatial dependence and censored measurements. A hierarchical GAM or mixed model could be used for that, with spatially blocked validation and sensitivity checks using alternative matches.

Connect the historical and current record

The same integration approach can reconnect a published historical cohort to current monitoring. ChemPop v2 identifies English river sites sampled through 2018. The open archive is downloaded from the EIDC record, then its site, sample and date identifiers are checked against the ecology archive acquired by this pipeline.

Dataset citation: Bachiller-Jareno, N.; Keller, V.D.J.; Eastman, M.; Edwards, F.K.; Jürgens, M.D.; Scarlett, P.M.; Rizzo, C.; Antoniou, V.; Dean, H.J.; Sadykova, D.; Qu, Y.; Johnson, A.C. (2025). *Macroinvertebrate taxonomic abundance, water quality, river flow, air temperature and environmental site descriptors from English rivers, 1965–2018, version 2.0*. NERC EDS Environmental Information Data Centre. DOI: 10.5285/23f194b0-1e21-4f14-be9c-6eabdcc1feb7.

Has macroinvertebrate condition changed, 1965–2025?

WHPT ASPT is taken from the same processed ecology archive throughout the time series, so the historical and recent values use one measure. The figures separate the number of sites sampled from the national trajectory and centre values within sites to reduce changes caused by the composition of the monitoring network.

The period comparison uses the same sites and calendar months in 1990–1999 and 2019–2025. It describes change in the observed record; it does not assign that change to a cause.

In [11]:
```
historical_archive = acquire_historical_archive(
    HISTORICAL_ARCHIVE_INPUT,
    WORKSPACE / "source_cache" / "historical_macroinvertebrates",
)
```

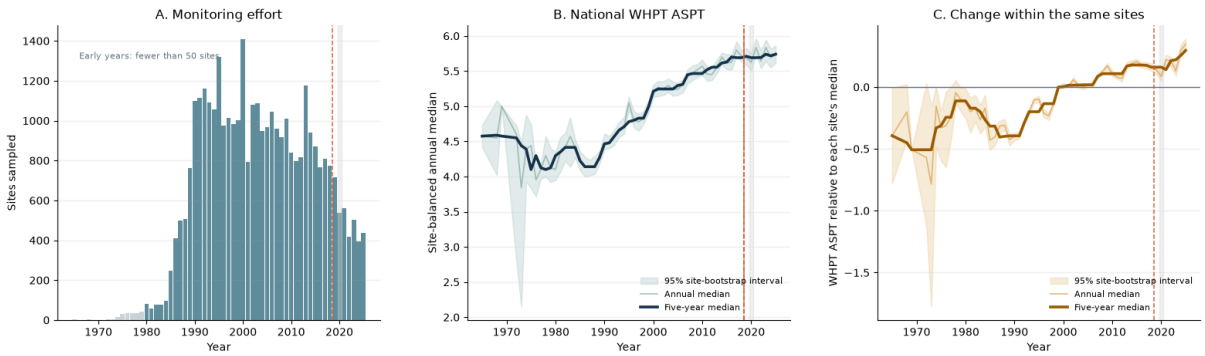
```
historical_result = build_historical_continuation(
    RELEASE_DIR, historical_archive, output_directory,
    historical_period=(1990, 1999),
)
display(historical_result.figure)

bridge = historical_result.bridge.iloc[0]
display(Markdown(
    "**Historical site matching.** "
    f"{int(bridge['exact samples found in current EA archive']):,} of "
    f"{int(bridge['deposited samples']):,} deposited samples match exactly "
    f"({bridge['exact match (%)']:.2f}%); "
    f"{int(bridge['sites observed after 2018']):,} of "
    f"{int(bridge['deposited sites']):,} original sites have usable later "
    "observations."
))

paired_display = historical_result.headline.copy()
paired_display["95% interval"] = paired_display.apply(
    lambda row: f"{row['lower 95%']:+.2f} to {row['upper 95%']:+.2f}", axis=
)
paired_display = paired_display[[
    "historical period", "recent period", "paired sites",
    "median within-site change", "95% interval",
    "sites with higher recent value (%)",
]].rename(columns={
    "median within-site change": "median change",
    "sites with higher recent value (%)": "sites higher (%)",
})
paired_display["paired sites"] = paired_display["paired sites"].map(
    lambda value: f"{int(value):,}"
)
paired_display["median change"] = paired_display["median change"].round(2)
paired_display["sites higher (%)"] = paired_display["sites higher (%)"].rour
display(Markdown("**Paired long-term comparison**"))
display(Markdown(dataframe_markdown(paired_display, include_index=False)))

display(Markdown(historical_result.summary_markdown))
```

Monitoring effort and WHPT ASPT through time



Historical site matching. 67,485 of 67,499 deposited samples match exactly (99.98%); 921 of 1,519 original sites have usable later observations.

Paired long-term comparison

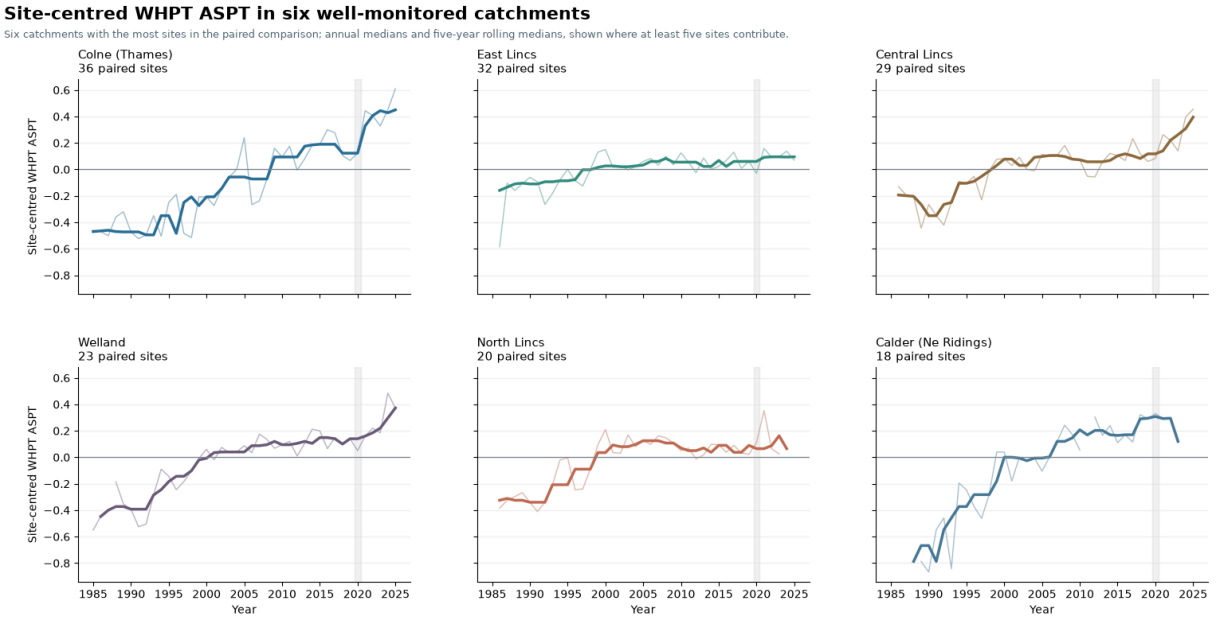
historical period	recent period	paired sites	median change	95% interval	sites higher (%)
1990–1999	2019–2025	848	0.52	+0.47 to +0.56	87.9

Long-term pattern. Coverage is sparse before the mid-1980s, so those early values are shown but not used for the period comparison. The five-year median site-centred value changes from -0.32 in 1985 to +0.29 in 2025. At hydrometric-area level, 33 of 34 areas with at least five paired sites have a positive median change. These are descriptive patterns and do not identify the cause of change.

Catchment trajectories

The six panels use the catchments with the most sites in the paired comparison. Lines are shown only when at least five sites contribute to a year.

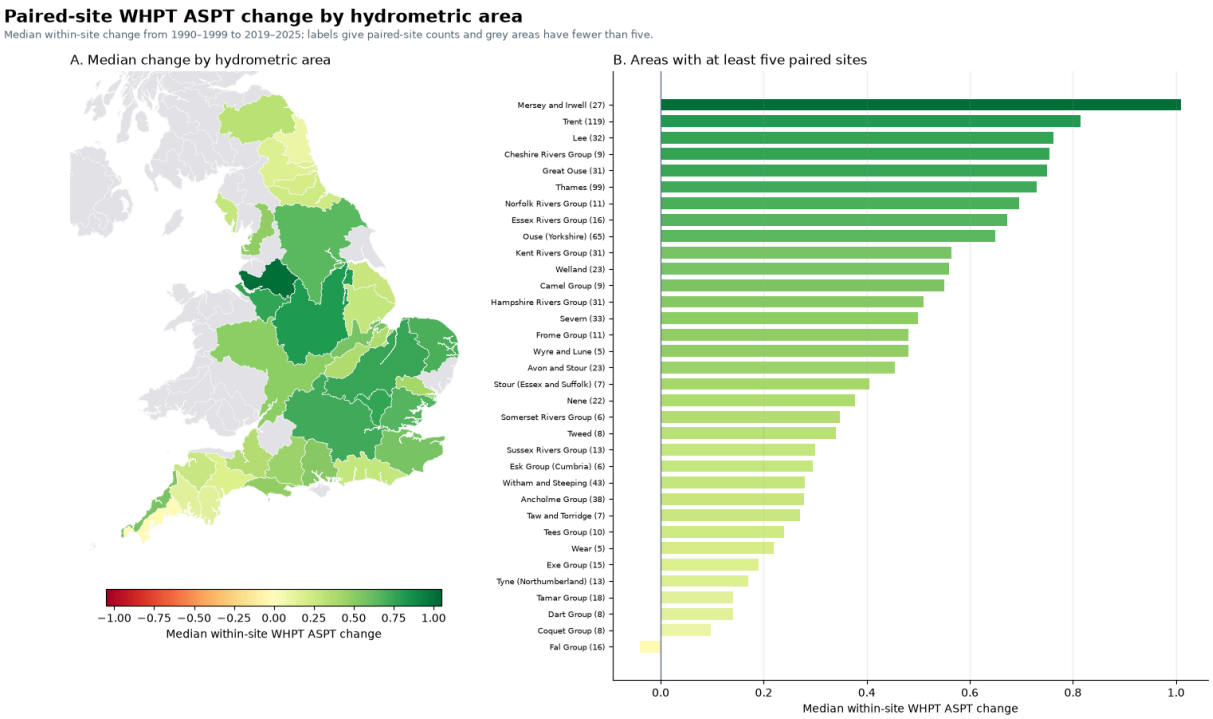
In [12]:
```
display(historical_result.catchment_figure)
```



Change by hydrometric area

The map and ranking summarise the same paired-site changes at a broader spatial scale. Areas with fewer than five paired sites remain grey.

In [13]:
```
display(historical_result.area_figure)
```



What did connecting the data enable?

The connected dataset shows where monitoring evidence overlaps and where gaps still limit analysis. It separates changes in monitoring effort from biological patterns and supports both national pressure comparisons and repeat-site change.

Because the integration is reusable, the matched evidence can support further questions without rebuilding the data connection. The Connected River (Part 1) - Building documents the method and its quality checks; this notebook demonstrates the analyses it enables.